

# The $10 \times 42$ Chomp bar has three winning moves

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## Abstract

In Chomp on an  $a \times b$  chocolate bar, Gale’s strategy-stealing argument shows the first player wins, but exhibits no winning move; famously, winning moves are known only by exhaustive computation, and their number varies. Ken Thompson found in the 1970s that the  $8 \times 10$  bar has two winning moves; Ekhad and Zeilberger (2018), tabulating all bars up to  $14 \times 14$ , found further doubles and challenged: find a bar with at least *three* winning moves. We answer the challenge: **the  $10 \times 42$  bar has exactly three winning bites, at  $(5, 36)$ ,  $(7, 30)$  and  $(8, 26)$ .** The computation is an exhaustive retrograde solve of the full ideal lattice, verified by two independent implementations and against the published tables. We also extend the winning-move-multiplicity census far beyond the 2018 range (eighteen bars with exactly two winning moves in the searched region), prove a small structural theorem — the winning bites of any bar form a strict staircase — and report striking quasi-periodic structure in where multiple winning moves occur, including bounded-error Beatty-type laws with rotation numbers in  $\mathbb{Q}(\sqrt{2})$  for the diagonal families of three-row Chomp.

## 1 The challenge

Chomp is played on an  $a \times b$  grid of cells (rows indexed downward); a move (a *bite*) at cell  $(i, j)$  removes every cell  $(r, c)$  with  $r \geq i$  and  $c \geq j$ ; the player forced to take the poisoned cell  $(1, 1)$  loses. Positions are exactly the Young-diagram ideals  $\lambda = (\lambda_1 \geq \dots \geq \lambda_a)$ . Gale’s strategy-stealing argument [1] shows every rectangular bar (other than  $1 \times 1$ ) is a first-player win, but famously certifies no winning move. Computation is the only known access, and already Gale asked whether the winning first move is always *unique*. The answer is no — Ken Thompson discovered that the  $8 \times 10$  bar has two winning moves (see [3]) — but the uniqueness phenomenon is real in low rows: Sheiner [4] has recently proved that every  $3 \times n$  bar has exactly one winning move, settling the three-row case of Gale’s question, and Garg [5] has tabulated the  $4 \times n$  P-positions to  $n \leq 500$ . Ekhad and Zeilberger [2], computing all bars up to  $14 \times 14$ , found a handful of doubles beyond Thompson’s  $(6 \times 13, 9 \times 10, 10 \times 14, 12 \times 13)$  and posed as their Second Computational Challenge: *find  $a, b$  such that the  $a \times b$  bar has at least three winning moves*. This note answers the challenge and maps the anti-uniqueness landscape.

**Answer.** The  $10 \times 42$  bar has exactly three winning bites: at  $(5, 36)$ ,  $(7, 30)$ , and  $(8, 26)$ .

## 2 A staircase theorem

**Theorem 1.** *If  $(i, j)$  and  $(i', j')$  are both winning bites of the same bar and  $i < i'$ , then  $j > j'$ . Moreover no winning bite of an  $a \times b$  bar with  $a \geq 2$ ,  $ab \geq 2$  lies in row 1.*

*Proof.* The bite  $(i, j)$  on the full bar leaves the two-block position  $B(i, j)$  with rows  $1, \dots, i-1$  of length  $b$  and rows  $i, \dots, a$  of length  $j-1$ . If  $i < i'$  and  $j \leq j'$  then biting  $B(i', j')$  at  $(i, j)$  yields exactly  $B(i, j)$ : two winning bites leave P-positions, and no P-position is one move from another. For the second claim, a row-1 bite leaves a full  $a \times (j-1)$  rectangle, an N-position by Gale (or, for  $j = 2$ , a column, which is N for  $a \geq 2$ ).  $\square$

The three bites of  $10 \times 42$  — rows  $5 < 7 < 8$ , columns  $36 > 30 > 26$  — form the first three-step staircase ever observed. Empirically the staircase is usually *tight* — in sixteen of the eighteen known doubles, and in the triple, consecutive winning bites are 1 or 2 rows apart — but not always: the two side-16 doubles split by six rows (Table 1).

## 3 Algorithm

One retrograde solve of the ideal lattice of an  $A \times B$  box determines the winning-move count of *every* bar  $a \times b$  with  $a \leq A$ ,  $b \leq B$ , since each such bar is a position in the box. We rank positions by the combinatorial number system on  $d_k = \lambda_k + A - k$  (strictly decreasing), i.e.  $\text{rank}(\lambda) = \sum_k \binom{d_k}{A-k+1}$ : this bijection onto  $\{0, \dots, \binom{A+B}{A} - 1\}$  is monotone under componentwise order, so a single increasing-rank sweep is a valid dynamic-programming order. The children of  $\lambda$  under a bite in row  $i$  at threshold  $t$  (all rows  $\geq i$  capped at  $t$ ) have ranks computable in  $O(1)$  amortized via the hockey-stick identity  $\sum_{m=m_1}^{m_2} \binom{t-1+m}{t-1} = \binom{t+m_2}{t} - \binom{t+m_1-1}{t}$  applied to the capped rows. The production engine stores one bit per position (P/N), detects N-positions with early exit at the first winning move, recovers exact counts for the bars in a final pass, and parallelizes by area layers (all positions of area  $s$  depend only on smaller areas). On 13 cores it solves the  $\binom{55}{10} \approx 2.9 \times 10^{10}$ -state  $10 \times 45$  box — the box that found the triple — in under half an hour.

## 4 Validation

(i) All engines agree with brute-force game search on six small boxes, both per-bar winning-move counts and the full P-position census. (ii) The published record is reproduced exactly: Thompson’s  $8 \times 10$  has two winning moves, at  $(4, 9)$  and  $(5, 6)$ , and the Ekhad–Zeilberger doubles  $6 \times 13$ ,  $10 \times 14$  check. (iii) The triple was found in the  $10 \times 45$  box, re-derived from the (different)  $10 \times 42$  box, and counter-signed by an independent implementation with a different iteration order and no early exit. All code and logs: <https://github.com/jaideepsaipadhi/chomp-three-winning-moves>.

## 5 The census

In the searched region — all  $a \leq b \leq 19$ ;  $4 \times b$ ,  $b \leq 300$ ;  $5 \times b$ ,  $b \leq 150$ ;  $6, 7 \times b$ ,  $b \leq 110$ ;  $8 \times b$ ,  $b \leq 60$ ;  $9 \times b$ ,  $b \leq 55$ ;  $10 \times b$ ,  $b \leq 45$  — there are exactly **eighteen** bars with exactly two winning moves and **one** with three:

$$6 \times 13, 6 \times 93, 7 \times 29, 7 \times 30, 7 \times 57, 8 \times 10, 8 \times 22, 8 \times 23, 9 \times 10, \\ 9 \times 26, 10 \times 14, 10 \times 29, 10 \times 33, 10 \times 35, 11 \times 18, 12 \times 13, 13 \times 16, 14 \times 16$$

(and transposes); every other bar in the region, including every square through  $19 \times 19$ , has a unique winning move; in particular  $4 \times b$  and  $5 \times b$  bars are unique throughout their ranges, consistent with (and, for four rows, implicit in the data of) Garg's tabulation [5]. Three regularities stand out. *Adjacent pairs*: doubles arrive at consecutive widths ( $7 \times 29/30$ ,  $8 \times 22/23$ ) or consecutive heights ( $13 \times 16$ ,  $14 \times 16$ ). *Hot and barren lines*: no bar with a side of 15 has a multiple move, while side 16 carries two doubles; row 10 carries six doubles *and* the triple. *Long gaps*: row 6 is silent from 14 to 92, then hits at 93.

Table 1: Winning bites of every known multi-move bar (staircase form guaranteed by Theorem 1); the crossing families  $(p, q) = (i-1, a-i+1)$  can be read off the bite rows.

| bar           | bites              | bar                              | bites                                         |
|---------------|--------------------|----------------------------------|-----------------------------------------------|
| $6 \times 13$ | $(4, 12), (5, 9)$  | $10 \times 14$                   | $(5, 13), (7, 9)$                             |
| $6 \times 93$ | $(2, 52), (3, 39)$ | $10 \times 29$                   | $(6, 16), (8, 12)$                            |
| $7 \times 29$ | $(6, 25), (7, 20)$ | $10 \times 33$                   | $(7, 23), (9, 18)$                            |
| $7 \times 30$ | $(4, 21), (5, 20)$ | $10 \times 35$                   | $(7, 24), (8, 21)$                            |
| $7 \times 57$ | $(3, 42), (4, 26)$ | $11 \times 18$                   | $(9, 17), (10, 16)$                           |
| $8 \times 10$ | $(4, 9), (5, 6)$   | $12 \times 13$                   | $(10, 11), (11, 9)$                           |
| $8 \times 22$ | $(2, 9), (3, 7)$   | $13 \times 16$                   | $(5, 6), (11, 5)$                             |
| $8 \times 23$ | $(7, 18), (8, 16)$ | $14 \times 16$                   | $(3, 12), (9, 10)$                            |
| $9 \times 10$ | $(7, 8), (9, 5)$   | <b><math>10 \times 42</math></b> | <b><math>(5, 36), (7, 30), (8, 26)</math></b> |
| $9 \times 26$ | $(2, 11), (4, 10)$ |                                  |                                               |

## 6 The structure of multiplicity

The count decomposes: biting the  $a \times b$  bar at  $(i, j)$  leaves the two-block position with  $p = i - 1$  full rows and  $q = a - p$  rows of length  $t = j - 1$ , so the number of winning moves of the bar equals the number of families  $(p, q)$ ,  $p + q = a$ , whose *P-track*  $\{(w, t) : (w^p, t^q) \text{ is a P-position}\}$  passes through width  $w = b$ . Empirically each track is sparse and quasi-linear — essentially one P-cell per width — so doubles are crossings of two tracks and the triple is a threefold crossing, constrained by Theorem 1 to a staircase. The two 7-row doubles, for instance, are the crossings  $(5, 2)-(6, 1)$  at  $b = 29$  and  $(3, 4)-(4, 3)$  at  $b = 30$ .

For three-row Chomp the tracks obey laws of unexpected exactness. Writing the  $n$ -th P-cell of the family  $(2, 1)$  (positions  $(w, w, t)$ ) as  $(w_n, t_n)$ , we find, over 247 cells up to  $w = 600$ ,

$$w_n = (1 + \sqrt{2})n + O(1), \quad t_n = \frac{2+\sqrt{2}}{2}n + O(1),$$

with error confined to a window of width  $< 3$  and *no drift*; the family  $(1, 2)$  (positions  $(w, t, t)$ ) hits essentially every  $t$ , with  $w(t) = \frac{2+\sqrt{2}}{2}t + O(1)$ , window  $< 2.4$  over 351 cells. The slopes' continued fractions match  $\sqrt{2} = [1; \overline{2}]$  and  $\frac{2+\sqrt{2}}{2}$  through all computed partial quotients. We state these as conjectures: the diagonal families of three-row Chomp are quasi-periodic with rotation numbers in  $\mathbb{Q}(\sqrt{2})$ . The exact three-row machinery of Brouwer et al. [3], in which Sheiner's uniqueness theorem [4] is proved, is the natural arena in which to attack them. This is

finer than, and consistent with, the renormalization picture of Friedman and Landsberg [7], who analyzed the three-row P-region probabilistically; Zeilberger’s assessment of three-row Chomp as chaotic [6] referred to the absence of eventual periodicity, which bounded-error quasi-periodicity with irrational rotation number precisely explains. For the higher families that govern the census above, the tracks appear to be multi-branch and their arithmetic remains open; the adjacent-pair phenomenon is *not* a single crossing lingering: the bite rows show that adjacent doubles arise from different family pairs ( $7 \times 29$  crosses  $(5, 2) - (6, 1)$  while  $7 \times 30$  crosses  $(3, 4) - (4, 3)$ ;  $8 \times 22$  crosses  $(1, 7) - (2, 6)$  while  $8 \times 23$  crosses  $(6, 2) - (7, 1)$ ), so the clustering of crossings at adjacent widths across *distinct* track pairs is itself the phenomenon demanding explanation. Whether triples occur infinitely often — and where the next one is — now looks like a question about simultaneous approximation of these rotation numbers.

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